

## Chapter 2.1

### Class 1 – Explosives

#### 2.1.0 Introductory notes (these notes are not mandatory)

- Note 1:** Class 1 is a restricted class, that is, only those explosive substances and articles that are listed in the Dangerous Goods List in chapter 3.2 may be accepted for transport. However, the competent authorities retain the right by mutual agreement to approve transport of explosive substances and articles for special purposes under special conditions. Therefore entries have been included in the Dangerous Goods List for “Substances, explosive, not otherwise specified” and “Articles, explosive, not otherwise specified”. It is intended that these entries should only be used when no other method of operation is possible.
- Note 2:** General entries such as “Explosive, blasting, type A” are used to allow for the transport of new substances. In preparing these provisions, military ammunition and explosives have been taken into consideration to the extent that they are likely to be transported by commercial carriers.
- Note 3:** A number of substances and articles in class 1 are described in appendix B. These descriptions are given because a term may not be well-known or may be at variance with its usage for regulatory purposes.
- Note 4:** Class 1 is unique in that the type of packaging frequently has a decisive effect on the hazard and therefore on the assignment to a particular division. The correct division is determined by use of the procedures provided in this chapter.

#### 2.1.1 Definitions and general provisions

##### 2.1.1.1 Class 1 comprises:

- .1 explosive substances (a substance which is not itself an explosive but which can form an explosive atmosphere of gas, vapour or dust is not included in class 1), except those which are too dangerous to transport or those where the predominant hazard is one appropriate to another class;
- .2 explosive articles, except devices containing explosive substances in such quantity or of such a character that their inadvertent or accidental ignition or initiation during transport shall not cause any effect external to the device either by projection, fire, smoke, heat or loud noise (see 2.1.3.4); and
- .3 substances and articles not mentioned under .1 and .2 which are manufactured with a view to producing a practical explosive or pyrotechnic effect.

##### 2.1.1.2 Transport of explosive substances which are unduly sensitive, or so reactive as to be subject to spontaneous reaction, is prohibited.

##### 2.1.1.3 Definitions

For the purposes of this Code, the following definitions apply:

- .1 *Explosive substance* means a solid or liquid substance (or a mixture of substances) which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings. Pyrotechnic substances are included even when they do not evolve gases.
- △ .2 *Pyrotechnic substance* means an explosive substance designed to produce an effect by heat, light, sound, gas or smoke or a combination of these as the result of non-detonative self-sustaining exothermic chemical reactions.
- .3 *Explosive article* means an article containing one or more explosive substances.
- .4 *Mass explosion* means one which affects almost the entire load virtually instantaneously.
- .5 *Phlegmatized* means that a substance (or “phlegmatizer”) has been added to an explosive to enhance its safety in handling and transport. The phlegmatizer renders the explosive insensitive, or less sensitive, to the following actions: heat, shock, impact, percussion or friction. Typical phlegmatizing agents include, but are not limited to: wax, paper, water, polymers (such as chlorofluoropolymers), alcohol and oils (such as petroleum jelly and paraffin).

- .6 *Explosive or pyrotechnic effect* means, in the context of 2.1.1.1.3, an effect produced by self-sustaining exothermic chemical reactions including shock, blast, fragmentation, projection, heat, light, sound, gas and smoke.

#### 2.1.1.4 Hazard divisions

The six hazard divisions of class 1 are:

Division 1.1 Substances and articles which have a mass explosion hazard

Division 1.2 Substances and articles which have a projection hazard but not a mass explosion hazard

Division 1.3 Substances and articles which have a fire hazard and either a minor blast hazard or a minor projection hazard or both, but not a mass explosion hazard

This division comprises substances and articles:

- .1 which give rise to considerable radiant heat; or
- .2 which burn one after another, producing minor blast or projection effects or both.

Division 1.4 Substances and articles which present no significant hazard

This division comprises substances and articles which present only a small hazard in the event of ignition or initiation during transport. The effects are largely confined to the package and no projection of fragments of appreciable size or range is to be expected. An external fire must not cause virtually instantaneous explosion of almost the entire contents of the package.

**Note:** Substances and articles in this division are in compatibility group S if they are so packaged or designed that any hazardous effects arising from the accidental functioning are confined within the package unless the package has been degraded by fire, in which case all blast or projection effects are limited to the extent that they do not significantly hinder fire fighting or other emergency response efforts in the immediate vicinity of the package.

Division 1.5 Very insensitive substances which have a mass explosion hazard

This division comprises substances which have a mass explosion hazard but are so insensitive that there is very little probability of initiation or of transition from burning to detonation under normal conditions of transport.

**Note:** The probability of transition from burning to detonation is greater when large quantities are transported in a ship. As a consequence, the stowage provisions for explosive substances in division 1.1 and for those in division 1.5 are identical.

Division 1.6 Extremely insensitive articles which do not have a mass explosion hazard

This division comprises articles which predominantly contain extremely insensitive substances and which demonstrate a negligible probability of accidental initiation or propagation.

**Note:** The hazard from articles of division 1.6 is limited to the explosion of a single article.

2.1.1.5 Any substance or article having or suspected of having explosive characteristics shall first be considered for classification in class 1 in accordance with the procedures in 2.1.3. Goods are not classified in class 1 when:

- .1 unless specially authorized, the transport of an explosive substance is prohibited because sensitivity of the substance is excessive;
- .2 the substance or article comes within the scope of those explosive substances and articles which are specifically excluded from class 1 by the definition of this class; or
- .3 the substance or article has no explosive properties.

#### 2.1.2 Compatibility groups and classification codes

2.1.2.1 Goods of class 1 are considered to be “compatible” if they can be safely stowed or transported together without significantly increasing either the probability of an accident or, for a given quantity, the magnitude of the effects of such an accident. By this criterion, goods listed in this class have been divided into a number of compatibility groups, each denoted by a letter from A to L (excluding I), N and S. These are described in 2.1.2.2 and 2.1.2.3.

## Part 2 – Classification

## 2.1.2.2 Compatibility groups and classification codes

| Description of substance or articles to be classified                                                                                                                                                                                                                                                                                                                                                          | Compatibility group | Classification code          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------|
| Primary explosive substance                                                                                                                                                                                                                                                                                                                                                                                    | A                   | 1.1A                         |
| Article containing a primary explosive substance and not containing two or more effective protective features. Some articles, such as detonators for blasting, detonator assemblies for blasting and primers, cap-type, are included even though they do not contain primary explosives                                                                                                                        | B                   | 1.1B<br>1.2B<br>1.4B         |
| Propellant explosive substance or other deflagrating explosive substance or article containing such explosive substance                                                                                                                                                                                                                                                                                        | C                   | 1.1C<br>1.2C<br>1.3C<br>1.4C |
| Secondary detonating explosive substance or black powder or article containing a secondary detonating explosive substance, in each case without means of initiation and without a propelling charge, or article containing a primary explosive substance and containing two or more effective protective features                                                                                              | D                   | 1.1D<br>1.2D<br>1.4D<br>1.5D |
| Article containing a secondary detonating explosive substance, without means of initiation, with a propelling charge (other than one containing a flammable liquid or gel or hypergolic liquids)                                                                                                                                                                                                               | E                   | 1.1E<br>1.2E<br>1.4E         |
| Article containing a secondary detonating explosive substance with its own means of initiation, with a propelling charge (other than one containing a flammable liquid or gel or hypergolic liquids) or without a propelling charge                                                                                                                                                                            | F                   | 1.1F<br>1.2F<br>1.3F<br>1.4F |
| Pyrotechnic substance, or article containing a pyrotechnic substance, or article containing both an explosive substance and an illuminating, incendiary, tear- or smoke-producing substance (other than a water-activated article or one containing white phosphorus, phosphides, a pyrophoric substance, a flammable liquid or gel, or hypergolic liquids)                                                    | G                   | 1.1G<br>1.2G<br>1.3G<br>1.4G |
| Article containing both an explosive substance and white phosphorus                                                                                                                                                                                                                                                                                                                                            | H                   | 1.2H<br>1.3H                 |
| Article containing both an explosive substance and a flammable liquid or gel                                                                                                                                                                                                                                                                                                                                   | J                   | 1.1J<br>1.2J<br>1.3J         |
| Article containing both an explosive substance and a toxic chemical agent                                                                                                                                                                                                                                                                                                                                      | K                   | 1.2K<br>1.3K                 |
| Explosive substance or article containing an explosive substance and presenting a special hazard (such as due to water-activation or presence of hypergolic liquids, phosphides or a pyrophoric substance) and needing isolation of each type (see 7.2.7.1.4, note 2)                                                                                                                                          | L                   | 1.1L<br>1.2L<br>1.3L         |
| Articles predominantly containing extremely insensitive substances                                                                                                                                                                                                                                                                                                                                             | N                   | 1.6N                         |
| Substance or article so packaged or designed that any hazardous effects arising from accidental functioning are confined within the package unless the package has been degraded by fire, in which case all blast or projection effects are limited to the extent that they do not significantly hinder or prohibit fire fighting or other emergency response efforts in the immediate vicinity of the package | S                   | 1.4S                         |

**Note 1:** Articles of compatibility groups D and E may be fitted or packed together with their own means of initiation provided that such means have at least two effective protective features designed to prevent an explosion in the event of accidental functioning of the means of initiation. Such articles and packages shall be assigned to compatibility groups D or E.

**Note 2:** Articles of compatibility groups D and E may be packed together with their own means of initiation, which do not have two effective protective features when, in the opinion of the competent authority of the country of origin, the accidental functioning of the means of initiation does not cause the explosion of an article under normal conditions of transport. Such packages shall be assigned to compatibility groups D or E.

**2.1.2.3 Scheme of classification of explosives, combination of hazard division with compatibility group**

| Hazard division | Compatibility group |      |      |      |      |      |      |      |      |      |      |      |      | Σ A–S |
|-----------------|---------------------|------|------|------|------|------|------|------|------|------|------|------|------|-------|
|                 | A                   | B    | C    | D    | E    | F    | G    | H    | J    | K    | L    | N    | S    |       |
| 1.1             | 1.1A                | 1.1B | 1.1C | 1.1D | 1.1E | 1.1F | 1.1G |      | 1.1J |      | 1.1L |      |      | 9     |
| 1.2             |                     | 1.2B | 1.2C | 1.2D | 1.2E | 1.2F | 1.2G | 1.2H | 1.2J | 1.2K | 1.2L |      |      | 10    |
| 1.3             |                     |      | 1.3C |      |      | 1.3F | 1.3G | 1.3H | 1.3J | 1.3K | 1.3L |      |      | 7     |
| 1.4             |                     | 1.4B | 1.4C | 1.4D | 1.4E | 1.4F | 1.4G |      |      |      |      |      | 1.4S | 7     |
| 1.5             |                     |      |      | 1.5D |      |      |      |      |      |      |      |      |      | 1     |
| 1.6             |                     |      |      |      |      |      |      |      |      |      |      | 1.6N |      | 1     |
| Σ 1.1–1.6       | 1                   | 3    | 4    | 4    | 3    | 4    | 4    | 2    | 3    | 2    | 3    | 1    | 1    | 35    |

**2.1.2.4** The definitions of compatibility groups in 2.1.2.2 are intended to be mutually exclusive, except for a substance or article which qualifies for compatibility group S. Since the criterion of compatibility group S is an empirical one, assignment to this group is necessarily linked to the tests for assignment to division 1.4.

**2.1.3 Classification procedure**

**2.1.3.1** Any substance or article having or suspected of having explosive characteristics shall be considered for classification in class 1. Substances and articles classified in class 1 shall be assigned to the appropriate division and compatibility group. Goods of class 1 shall be classified in accordance with the *Manual of Tests and Criteria*, as amended.

**2.1.3.2** Prior to transport, the classification of all explosive substances and articles, together with the compatibility group assignment and the proper shipping name under which the substance or article is to be transported, shall have been approved by the competent authority of the country of manufacture. A new approval would be required for:

- .1 a new explosive substance; or
- .2 a new combination or mixture of explosive substances which is significantly different from other combinations or mixtures previously manufactured and approved; or
- .3 a new design of an explosive article, an article containing a new explosive substance, or an article containing a new combination or mixture of explosive substances; or
- .4 an explosive substance or article with a new design or type of packaging, including a new type of inner packaging.

**2.1.3.3** Assessment of the hazard division is usually made on the basis of test results. A substance or article shall be assigned to the hazard division which corresponds to the results of the tests to which the substance or article, as offered for transport, has been subjected. Other test results, and data assembled from accidents which have occurred, may also be taken into account.

**2.1.3.4 Exclusion from class 1**

**2.1.3.4.1** The competent authority may exclude an article or substance from class 1 by virtue of test results and the class 1 definition.

**2.1.3.4.2** An article may be excluded from class 1 by the competent authority when three unpackaged articles, each individually activated by its own means of initiation or ignition or external means to function in the designed mode, meet the following test criteria:

- .1 no external surface shall have a temperature of more than 65°C. A momentary spike in temperature up to 200°C is acceptable;
- .2 no rupture or fragmentation of the external casing or movement of the article or detached parts thereof of more than one metre in any direction;

**Note:** Where the integrity of the article may be affected in the event of an external fire these criteria shall be examined by a fire test. One such method is described in ISO 14451-2 using a heating rate of 80 K/min.

- .3 no audible report exceeding 135 dB(C) peak at a distance of one metre;
- .4 no flash or flame capable of igniting a material such as a sheet of 80 ± 10 g/m<sup>2</sup> paper in contact with the article; and

- .5 no production of smoke, fumes or dust in such quantities that the visibility in a one cubic metre chamber equipped with appropriately sized blow out panels is reduced more than 50% as measured by a calibrated light (lux) meter or radiometer located one metre from a constant light source located at the midpoint on opposite walls. The general guidance on Optical Density Testing in ISO 5659-1 and the general guidance on the Photometric System described in Section 7.5 in ISO 5659-2 may be used or similar optical density measurement methods designed to accomplish the same purpose may also be employed. A suitable hood cover surrounding the back and sides of the light meter shall be used to minimize effects of scattered or leaking light not emitted directly from the source.

**Note 1:** If during the tests addressing criteria .1, .2, .3 and .4 no or very little smoke is observed the test described in .5 may be waived.

**Note 2:** The competent authority may require testing in packaged form if it is determined that, as packaged for transport, the article may pose a greater hazard.

### 2.1.3.5 Assignment of fireworks to hazard divisions

2.1.3.5.1 Fireworks shall normally be assigned to hazard divisions 1.1, 1.2, 1.3, and 1.4 on the basis of test data derived from Test Series 6 of the *Manual of Tests and Criteria*. However:

- .1 waterfalls containing flash composition (see note 2 of 2.1.3.5.5) shall be classified as 1.1G regardless of the results of Test Series 6;
- .2 since the range of fireworks is very extensive and the availability of test facilities may be limited, assignment to hazard divisions may also be made in accordance with the procedure in 2.1.3.5.2.

2.1.3.5.2 Assignment of fireworks to UN Nos. 0333, 0334, 0335 or 0336, and assignments of articles to UN 0431 for those used for theatrical effects meeting the definition for article type and the 1.4G specification in the default fireworks classification table in 2.1.3.5.5 may be made on the basis of analogy, without the need for Test Series 6 testing, in accordance with the default fireworks classification table in 2.1.3.5.5. Such assignment shall be made with the agreement of the competent authority. Items not specified in the table shall be classified on the basis of test data derived from Test Series 6 of the *Manual of Tests and Criteria*.

**Note:** The addition of other types of fireworks to column 1 of the table in 2.1.3.5.5 shall only be made on the basis of full test data submitted to the UN Sub-Committee of Experts on the Transport of Dangerous Goods for consideration.

2.1.3.5.3 Where fireworks of more than one hazard division are packed in the same package they shall be classified on the basis of the highest hazard division unless test data derived from Test Series 6 of the *Manual of Tests and Criteria* indicate otherwise.

2.1.3.5.4 The classification shown in the table in 2.1.3.5.5 applies only for articles packed in fibreboard boxes (4G).

### 2.1.3.5.5 Default fireworks classification table\*

**Note 1:** References to percentages in the table, unless otherwise stated, are to the mass of all pyrotechnic substances (e.g. rocket motors, lifting charge, bursting charge and effect charge).

**Note 2:** "Flash composition" in this table refers to pyrotechnic substances in powder form or as pyrotechnic units as presented in the fireworks that are used in waterfalls, or to produce an aural effect or used as a bursting charge, or propellant charge unless:

- (a) the time taken for the pressure rise in the HSL Flash Composition Test in appendix 7 of the *Manual of Tests and Criteria* is demonstrated to be more than 6 ms for 0.5 g of pyrotechnic substance; or
- (b) the pyrotechnic substance gives a negative "–" result in the US Flash Composition Test in Appendix 7 of the *Manual of Tests and Criteria*.

**Note 3:** Dimensions in millimetres refers to:

- for spherical and peanut shells, the diameter of the sphere of the shell;
- for cylinder shells, the length of the shell;
- for a shell in mortar, Roman candle, shot tube firework or mine, the inside diameter of the tube comprising or containing the firework;
- for a bag mine or cylinder mine, the inside diameter of the mortar intended to contain the mine.

\*This table contains a list of firework classifications that may be used in the absence of Test Series 6, of the *Manual of Tests and Criteria*, data (see 2.1.3.5.2).

| Type                            | Includes: / Synonym:                                                                                                                                                                                                                      | Definition                                                                                                                                                                   | Specification                                                                                                                              | Classification |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Shell, spherical or cylindrical | Spherical display shell: aerial shell, colour shell, dye shell, multi-break shell, multi-effect shell, nautical shell, parachute shell, smoke shell, star shell; report shell: maroon, salute, sound shell, thunderclap, aerial shell kit | Device with or without propellant charge, with delay fuse and bursting charge, pyrotechnic unit(s) or loose pyrotechnic substance and designed to be projected from a mortar | All report shells                                                                                                                          | 1.1G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $\geq 180$ mm                                                                                                                | 1.1G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $< 180$ mm with $> 25\%$ flash composition, as loose powder and/or report effects                                            | 1.1G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $< 180$ mm with $\leq 25\%$ flash composition, as loose powder and/or report effects                                         | 1.3G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $\leq 50$ mm, or $\leq 60$ g pyrotechnic substance, with $\leq 2\%$ flash composition as loose powder and/or report effects  | 1.4G           |
|                                 | Peanut shell                                                                                                                                                                                                                              | Device with two or more spherical aerial shells in a common wrapper propelled by the same propellant charge with separate external delay fuses                               | The most hazardous spherical aerial shell determines the classification                                                                    |                |
|                                 | Preloaded mortar, shell in mortar                                                                                                                                                                                                         | Assembly comprising a spherical or cylindrical shell inside a mortar from which the shell is designed to be projected                                                        | All report shells                                                                                                                          |                |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $\geq 180$ mm                                                                                                                | 1.1G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $> 25\%$ flash composition as loose powder and/or report effects                                                             | 1.1G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $> 50$ mm and $< 180$ mm                                                                                                     | 1.2G           |
|                                 |                                                                                                                                                                                                                                           |                                                                                                                                                                              | Colour shell: $\leq 50$ mm, or $\leq 60$ g pyrotechnic substance, with $\leq 25\%$ flash composition as loose powder and/or report effects | 1.3G           |

## Part 2 – Classification

| Type                    | Includes: / Synonym:                                                                                                               | Definition                                                                                                                                                                                                                                  | Specification                                                                                 | Classification |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------|
| Battery/<br>combination | Shell of shells (spherical)<br>(Reference to percentages for shell<br>of shells are to the gross mass of the<br>fireworks article) | Device without propellant charge, with delay fuse and bursting charge, containing report shells and inert materials and designed to be projected from a mortar                                                                              | > 120 mm                                                                                      | 1.1G           |
|                         |                                                                                                                                    | Device without propellant charge, with delay fuse and bursting charge, containing report shells ≤ 25 g flash composition per report unit, with ≤ 33% flash composition and ≥ 60% inert materials and designed to be projected from a mortar | ≤ 120 mm                                                                                      | 1.3G           |
|                         |                                                                                                                                    | Device without propellant charge, with delay fuse and bursting charge, containing colour shells and/or pyrotechnic units and designed to be projected from a mortar                                                                         | > 300 mm                                                                                      | 1.1G           |
|                         |                                                                                                                                    | Device without propellant charge, with delay fuse and bursting charge, containing colour shells ≤ 70 mm and/or pyrotechnic units, with ≤ 25% flash composition and ≤ 60% pyrotechnic substance and designed to be projected from a mortar   | > 200 mm and ≤ 300 mm                                                                         | 1.3G           |
|                         |                                                                                                                                    | Device with propellant charge, with delay fuse and bursting charge, containing colour shells ≤ 70 mm and/or pyrotechnic units, with ≤ 25% flash composition and ≤ 60% pyrotechnic substance and designed to be projected from a mortar      | ≤ 200 mm                                                                                      | 1.3G           |
| Roman candle            | Barrage, bombardos, cakes, finale box, flowerbed, hybrid, multiple tubes, shell cakes, banger batteries, flash banger batteries    | Assembly including several elements either containing the same type or several types each corresponding to one of the types of fireworks listed in this table, with one or two points of ignition                                           | The most hazardous firework type determines the classification                                |                |
|                         |                                                                                                                                    | Tube containing a series of pyrotechnic units consisting of alternate pyrotechnic substance, propellant charge, and transmitting fuse                                                                                                       | ≥ 50 mm inner diameter, containing flash composition, or < 50 mm with > 25% flash composition | 1.1G           |
|                         |                                                                                                                                    |                                                                                                                                                                                                                                             | ≥ 50 mm inner diameter, containing no flash composition                                       | 1.2G           |
|                         |                                                                                                                                    |                                                                                                                                                                                                                                             | < 50 mm inner diameter and ≤ 25% flash composition                                            | 1.3G           |
| Shot tube               | Single shot Roman candle, small preloaded mortar                                                                                   | Tube containing a pyrotechnic unit consisting of pyrotechnic substance, propellant charge with or without transmitting fuse                                                                                                                 | ≤ 30 mm inner diameter, each pyrotechnic unit ≤ 25 g and ≤ 5% flash composition               | 1.4G           |
|                         |                                                                                                                                    |                                                                                                                                                                                                                                             | ≤ 30 mm inner diameter and pyrotechnic unit > 25 g, or > 5% and ≤ 25% flash composition       | 1.3G           |
|                         |                                                                                                                                    |                                                                                                                                                                                                                                             | ≤ 30 mm inner diameter, pyrotechnic unit ≤ 25 g and ≤ 5% flash composition                    | 1.4G           |



| Type         | Includes: / Synonym:                                                                                             | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Classification                                                   |
|--------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Rocket       | Avalanche rocket, signal rocket, whistling rocket, bottle rocket, sky rocket, missile type rocket, table rocket  | Tube containing pyrotechnic substance and/or pyrotechnic units, equipped with stick(s) or other means for stabilization of flight, and designed to be propelled into the air                                                                                                                                                                                                                                                                              | Flash composition effects only<br><br>Flash composition > 25% of the pyrotechnic substance<br><br>> 20 g pyrotechnic substance and flash composition ≤ 25 %<br><br>≤ 20 g pyrotechnic substance, black powder bursting charge and ≤ 0.13 g flash composition per report and ≤ 1 g in total                                                                                                                                                                                                                      | 1.1G<br><br>1.1G<br><br>1.3G<br><br>1.4G                         |
| Mine         | Pot-a-feu, ground mine, bag mine, cylinder mine                                                                  | Tube containing propellant charge and pyrotechnic units and designed to be placed on the ground or to be fixed in the ground. The principal effect is ejection of all the pyrotechnic units in a single burst producing a widely dispersed visual and/or aural effect in the air or:<br><br>Cloth or paper bag or cloth or paper cylinder containing propellant charge and pyrotechnic units, designed to be placed in a mortar and to function as a mine | > 25% flash composition, as loose powder and/or report effects<br><br>≥ 180 mm and ≤ 25% flash composition, as loose powder and/or report effects<br><br>< 180 mm and ≤ 25% flash composition, as loose powder and/or report effects<br><br>≤ 150 g pyrotechnic substance, containing ≤ 5% flash composition as loose powder and/or report effects. Each pyrotechnic unit ≤ 25 g, each report effect < 2 g; each whistle, if any, ≤ 3 g<br><br>≥ 1 kg pyrotechnic substance<br><br>< 1 kg pyrotechnic substance | 1.1G<br><br>1.1G<br><br>1.3G<br><br>1.4G<br><br>1.3G<br><br>1.4G |
| Fountain     | Volcanos, gerbs, lances, Bengal fire, flitter sparkle, cylindrical fountains, cone fountains, illuminating torch | Non-metallic case containing pressed or consolidated pyrotechnic substance producing sparks and flame<br><b>Note:</b> Fountains intended to produce a vertical cascade or curtain of sparks are considered to be waterfalls (see row below).                                                                                                                                                                                                              | Containing flash composition regardless of the results of Test Series 6 (see 2.1.3.5.1.1)                                                                                                                                                                                                                                                                                                                                                                                                                       | 1.1G                                                             |
| Waterfall    | Cascades, showers                                                                                                | Pyrotechnic fountain intended to produce a vertical cascade or curtain of sparks                                                                                                                                                                                                                                                                                                                                                                          | Not containing flash composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.3G                                                             |
| Sparkler     | Handheld sparklers, non-handheld sparklers, wire sparklers                                                       | Rigid wire partially coated (along one end) with slow-burning pyrotechnic substance with or without an ignition tip                                                                                                                                                                                                                                                                                                                                       | Perchlorate based sparklers: > 5 g per item or > 10 items per pack<br><br>Perchlorate based sparklers: ≤ 5 g per item and ≤ 10 items per pack<br>Nitrate based sparklers: ≤ 30 g per item                                                                                                                                                                                                                                                                                                                       | 1.3G<br><br>1.4G                                                 |
| Bengal stick | Dipped stick                                                                                                     | Non-metallic stick partially coated (along one end) with slow-burning pyrotechnic substance and designed to be held in the hand                                                                                                                                                                                                                                                                                                                           | Perchlorate based items: > 5 g per item or > 10 items per pack<br><br>Perchlorate based items: ≤ 5 g per item and ≤ 10 items per pack; nitrate based items: ≤ 30 g per item                                                                                                                                                                                                                                                                                                                                     | 1.3G<br><br>1.4G                                                 |



## Part 2 – Classification

| Type                               | Includes: / Synonym:                                                                                        | Definition                                                                                                                                                         | Specification                                                                                                                                                                                                                                                                                                                                                                                          | Classification       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Low hazard fireworks and novelties | Table bombs, throwdowns, crackling granules, smokes, fog, snakes, glow worm, serpents, snaps, party poppers | Device designed to produce very limited visible and/or audible effect which contains small amounts of pyrotechnic and/or explosive substance                       | Throwdowns and snaps may contain up to 1.6 mg of silver fulminate; snaps and party poppers may contain up to 16 mg of potassium chlorate/red phosphorus mixture; other articles may contain up to 5 g of pyrotechnic substance, but no flash composition                                                                                                                                               | 1.4G                 |
| Spinner                            | Aerial spinner, helicopter, chaser, ground spinner                                                          | Non-metallic tube or tubes containing gas- or spark-producing pyrotechnic substance, with or without noise-producing substance, with or without aerofoils attached | Pyrotechnic substance per item > 20 g, containing ≤ 3% flash composition as report effects, or whistle composition ≤ 5 g<br><br>Pyrotechnic substance per item ≤ 20 g, containing ≤ 3% flash composition as report effects, or whistle composition ≤ 5 g                                                                                                                                               | 1.3G<br><br>1.4G     |
| Wheels                             | Catherine wheels, Saxon                                                                                     | Assembly including drivers containing pyrotechnic substance and provided with a means of attaching it to a support so that it can rotate                           | ≥ 1 kg total pyrotechnic substance, no report effect, each whistle (if any) ≤ 25 g and ≤ 50 g whistle composition per wheel<br><br>< 1 kg total pyrotechnic substance, no report effect, each whistle (if any) ≤ 5 g and ≤ 10 g whistle composition per wheel                                                                                                                                          | 1.3G<br><br>1.4G     |
| Aerial wheel                       | Flying Saxon, UFOs, rising crown                                                                            | Tubes containing propellant charges and sparks-, flame- and/or noise-producing pyrotechnic substances, the tubes being fixed to a supporting ring                  | > 200 g total pyrotechnic substance or > 60 g pyrotechnic substance per driver, ≤ 3% flash composition as report effects, each whistle (if any) ≤ 25 g and ≤ 50 g whistle composition per wheel<br><br>≤ 200 g total pyrotechnic substance and ≤ 60 g pyrotechnic substance per driver, ≤ 3% flash composition as report effects, each whistle (if any) ≤ 5 g and ≤ 10 g whistle composition per wheel | 1.3G<br><br>1.4G     |
| Selection pack                     | Display selection box, display selection pack, garden selection box, indoor selection box; assortment       | A pack of more than one type each corresponding to one of the types of fireworks listed in this table                                                              | The most hazardous fireworks type determines the classification                                                                                                                                                                                                                                                                                                                                        |                      |
| Firecracker                        | Celebration cracker, celebration roll, string cracker                                                       | Assembly of tubes (paper or cardboard) linked by a pyrotechnic fuse, each tube intended to produce an aural effect                                                 | Each tube ≤ 140 mg of flash composition or ≤ 1 g black powder                                                                                                                                                                                                                                                                                                                                          | 1.4G                 |
| Banger                             | Salute, flash banger, lady cracker                                                                          | Non-metallic tube containing report composition intended to produce an aural effect                                                                                | > 2 g flash composition per item<br>≤ 2 g flash composition per item and ≤ 10 g per inner packaging<br>≤ 1 g flash composition per item and ≤ 10 g per inner packaging or ≤ 10 g black powder per item                                                                                                                                                                                                 | 1.1G<br>1.3G<br>1.4G |

### 2.1.3.6 Classification documentation

2.1.3.6.1 A competent authority assigning an article or substance into class 1 should confirm with the applicant that classification in writing.

2.1.3.6.2 A competent authority classification document may be in any form and may consist of more than one page, provided pages are numbered consecutively. The document should have a unique reference.

2.1.3.6.3 The information provided shall be easy to identify, legible and durable.

2.1.3.6.4 Examples of the information that may be provided in the classification documents are as follows:

- .1 the name of the competent authority and the provisions in national legislation under which it is granted its authority;
- .2 the modal or national regulations for which the classification document is applicable;
- .3 confirmation that the classification has been approved, made or agreed in accordance with the United Nations Recommendations on the Transport of Dangerous Goods or the relevant modal regulations;
- .4 the name and address of the person in law to which the classification has been assigned and any company registration which uniquely identifies a company or other body corporate under national legislation;
- .5 the name under which the explosives will be placed on the market or otherwise supplied for transport;
- .6 the Proper Shipping Name, UN number, Class, Hazard Division and corresponding compatibility group of the explosives;
- .7 where appropriate, the maximum net explosive mass of the package or article;
- .8 the name, signature, stamp, seal or other identification of the person authorized by the competent authority to issue the classification document is clearly visible;
- .9 where safety in transport or the hazard division is assessed as being dependent upon the packaging, the packaging mark or a description of the permitted:
  - inner packagings
  - intermediate packagings
  - outer packagings
- .10 the classification document states the part number, stock number or other identifying reference under which the explosives will be placed onto the market or otherwise supplied for transport;
- .11 the name and address of the person in law who manufactured the explosives and any company registration which uniquely identifies a company or other body corporate under national legislation;
- .12 any additional information regarding the applicable packing instruction and special packing provisions where appropriate;
- .13 the basis for assigning the classification, i.e. whether on the basis of test results, default for fireworks, analogy with classified explosive, by definition from the Dangerous Goods List etc.;
- .14 any special conditions or limitations that the competent authority has identified as relevant to the safety for transport of the explosives, the communication of the hazard and international transport; and
- .15 the expiry date of the classification document is given where the competent authority considers one to be appropriate.